SUPPORTING ADULT GUIDANCE

Deriving a formula

Formulae and Rearranging · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson asks learners to derive formulae for themselves and to justify them from structure rather than from a table alone. Learners find hidden rules from tables, build surface-area formulae face by face from nets, match four different groupings of a border pattern to four equivalent formulae, and use a pattern builder to colour what stays fixed and what grows before deriving, testing and explaining a formula — then design a pattern of their own.

Driving Question: *Can I find the rule myself, and say why it works?*

The mathematics, in plain terms

For a linear pattern, the term-to-term difference d gives the coefficient of n, and the constant is found from the first term: T = dn + (T■ – d). A structural derivation identifies these parts in the diagram, which justifies the formula for all n. Different structural groupings give algebraically equivalent expressions, for example $4n + 4 \equiv 4(n + 1) \equiv 2(n + 2) + 2n \equiv (n + 2)^2 - n^2$.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Squares made from rods

One square uses 4 rods, two squares in a row use 7, three use 10. Derive a formula for the number of rods R in step n.

1. What stays the same? The first rod on the left.	1
2. What grows? Each square adds 3 rods: top, bottom and right.	$3n$
3. Put them together.	$R = 3n + 1$
4. Test on step 50.	$3 \times 50 + 1 = 151$ rods

$R = 3n + 1$, because there is one starting rod and 3 more for each square.

Example 2 — The surface area of a cuboid

A cuboid has edges a, b and c. Derive a formula for its surface area S.

1. The front and back faces are each a × b.	2ab
2. The two sides are each a × c.	2ac
3. The top and bottom are each b × c.	2bc
4. Add the three pairs.	S = 2ab + 2ac + 2bc

S = 2ab + 2ac + 2bc. Each 2 is a pair of matching faces.

Point at the formula

The strongest evidence of understanding in this lesson is a learner pointing at the picture and saying which part is which term. Ask 'Show me the $3n$' and 'Show me the + 1'. A correct formula from the table alone is a good start, but the lesson's aim is the explanation.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The rule detective: find a hidden formula from a table, then test it on new inputs.	~6 min
Movement	Net to formula: tap each face of a cuboid's net to build its surface area formula.	~8 min
Reflection	Four ways of seeing one border: match groupings to equivalent formulae.	~10 min
Creativity	The pattern builder: colour the structure, derive, test on step 50, explain — then design your own.	~16 min
Rest	Step 50: too big to draw, already counted.	~2 min
Nutrition	Intellectual: knowing why something is always true — the first step towards proof.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Writing the term-to-term rule (n + 3) instead of the position-to-term rule (3n + 1).
Test it: does n + 3 give 4 when n = 1, and 7 when n = 2?

Getting the multiplier right but the constant wrong (3n + 4).
Ask the learner to check step 1 and to find the part that stays the same in the picture.

Counting shared rods or tiles twice.
Colour each rod once as you count it.

Thinking two different-looking formulae for the same pattern cannot both be right.
Compare them in a table for several values of n, then link to identities from Lesson 1.

Forgetting that a cuboid has two of each face.
Ask the learner to point to the face opposite each one on a real box.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: linear sequences and the nth term, and writing formulae from Lesson 3. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes n + 3 for the rods pattern 4, 7, 10. The aim is a question that lets them find the gap themselves.

- "What does your formula give for step 1? And for step 2?"
- "How many rods are added each time? Where is that in your formula?"

- “Can you colour the rods that are the same in every step?”
- “How many rods would step 50 need — without drawing it?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is a formula from the table alone enough?
It is correct, and accepted. The lesson then asks for the explanation from the picture, because that is what makes it a derivation rather than a guess that fits.

Why are there four ways of seeing the border?
To show that equivalent formulae come from different, equally valid ways of grouping — and to connect with identities from Lesson 1.

Why do the net and box use a, b and c?
To avoid the letter l, which can be mistaken for the number 1.

What comes next?
Lesson 5 changes the subject of a formula: turning it round to find what you need.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.

Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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